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<https://swayam.gov.in>https://swayam.gov.in/nc_details/NPTEL

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NPTEL (<https://swayam.gov.in/explorer?ncCode=NPTEL>) » Introduction To Machine Learning - IITKGP (course)



Course outline

About NPTEL ()

How does an NPTEL online course work? ()

Week 0 ()

Week 1 ()

- ☐ Lecture 01: Introduction (unit?unit=6&lesson=7)

- ☐ Lecture 02 : Different Types of Learning (unit?unit=6&lesson=8)

- ☒ Lecture 03 : Hypothesis Space and Inductive Bias (unit?

Week 1 : Assignment 1

The due date for submitting this assignment has passed.

Due on 2024-08-07, 23:59 IST.

Assignment submitted on 2024-08-07, 23:57 IST

1)

2 points

Which of the following is a classification task?

- A. Detect pneumonia from chest X-ray image
- B. Predict the price of a house based on floor area, number of rooms etc.
- C. Predict the temperature for the next day
- D. Predict the amount of rainfall

☒ A.

☐ B.

☐ C.

☐ D.

Yes, the answer is correct.

Score: 2

Accepted Answers:

A.

2)

2 points

Which of the following is not a type of supervised learning?

- A. Classification
- B. Regression
- C. Clustering
- D. None of the above

unit=6&lesson=9)

☐ Lecture 04 :
Evaluation and Cross-Validation
(unit? unit=6&lesson=10)

☐ Lecture 5: Tutorial - I
(unit? unit=6&lesson=11)

☐ Week 1 :
Lecture materials
(unit? unit=6&lesson=12)

☒ **Quiz: Week 1 : Assignment 1**
(assessment? name=142)

☐ Feedback Form for Week 1 (unit? unit=6&lesson=13)

☐ Assignment 1 Solution (unit? unit=6&lesson=120)

Week 2 ()

Week 3 ()

Week 4 ()

Week 5 ()

Week 6 ()

Week 7 ()

Week 8 ()

Assignment Solution ()

- ☐ A.
☐ B.
☒ C.
☐ D.

Yes, the answer is correct.

Score: 2

Accepted Answers:

C.

3)

2 points

Which of the following tasks is NOT a suitable machine learning task?

- A. Finding the shortest path between a pair of nodes in a graph
B. Predicting if a stock price will rise or fall
C. Predicting the price of petroleum
D. Grouping mails as spams or non-spams

- ☒ A.
☐ B.
☐ C.
☐ D.

Yes, the answer is correct.

Score: 2

Accepted Answers:

A.

4)

2 points

Suppose I have 10,000 emails in my mailbox out of which 300 are spams. The spam detection system detects 150 mails as spams, out of which 50 are actually spams. What is the precision and recall of my spam detection system?

- A. Precision = 33.33%, Recall = 25%
B. Precision = 25%, Recall = 33.33%
C. Precision = 33.33%, Recall = 16.66%
D. Precision = 75%, Recall = 33.33%

- ☐ A.
☐ B.
☒ C.
☐ D.

Yes, the answer is correct.

Score: 2

Accepted Answers:

C.

5)

2 points

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July 2024 ()

Which of the following is/are supervised learning problems?

- A. Predicting disease from blood samples.
- B. Grouping students in the same class based on similar features.
- C. Face recognition to unlock your phone.

- ☒ A.
- ☐ B.
- ☒ C.

Yes, the answer is correct.

Score: 2

Accepted Answers:

- A.
- C.

6)

2 po

Aliens challenge you to a complex game that no human has seen before. They give you time to learn the game and develop strategies before the final showdown. You choose to use machine learning because an intelligent machine is your only hope. Which machine learning paradigm should you choose for this?

- A. Supervised learning
- B. Unsupervised learning
- C. Reinforcement learning
- D. Use a random number generator and hope for the best

- ☐ A.
- ☐ B.
- ☒ C.
- ☐ D.

Yes, the answer is correct.

Score: 2

Accepted Answers:

- C.

7)

2 points

How many Boolean functions are possible with N features?

- A. (2^{2^N})
- B. (2^N)
- C. (N^2)
- D. (4^N)

- ☒ A.
- ☐ B.
- ☐ C.
- ☐ D.

Yes, the answer is correct.

Score: 2

Accepted Answers:

A.

8)

2 points

What is the use of Validation dataset in Machine Learning?

- A. To train the machine learning model.
- B. To evaluate the performance of the machine learning model
- C. To tune the hyperparameters of the machine learning model
- D. None of the above.

☐ A.

☐ B.

☒ C.

☐ D.

Yes, the answer is correct.

Score: 2

Accepted Answers:

C.

9)

2 points

Regarding bias and variance, which of the following statements are true? (Here 'high' and 'low' are relative to the ideal model.)

- A. Models which overfit have a high bias.
- B. Models which overfit have a low bias.
- C. Models which underfit have a high variance.
- D. Models which underfit have a low variance.

☐ A.

☒ B.

☐ C.

☒ D.

Yes, the answer is correct.

Score: 2

Accepted Answers:

B.

D.

10) Which of the following is a categorical feature?

2 points

- A. Height of a person
- B. Price of petroleum
- C. Mother tongue of a person
- D. Amount of rainfall in a day

- ☐ A.
- ☐ B.
- ☒ C.
- ☐ D.

Yes, the answer is correct.

Score: 2

Accepted Answers:

C.

